COFFEE SKILLS PROGRAM

Barista Skills

Professional AST Guidebook (中文)

Specialty Coffee Association

Barista Skills Professional

AST Guidebook V1.0（中文/Simplified Chinese）

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致謝：

過去十多年來，無數的咖啡專業人士和具有天賦的教育工作者為 SCA 認證教育計畫自願付出了他們的寶貴的時間和專業知識。以下列出最新的貢獻者，他們藉由貢獻以及驗證目前的證書課程教材來延續此傳承。

Ben Townsend, Benco Consulting
Paul Meikle-Janney, Dark Wood Coffee
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Miguel Vicuña, Sweet Bloom Coffee Roasters
Nick Choi, Europ Speciality Coffee Academy/Interobang Coffee Company
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注意：本指南替代了以前的课程文件。本指南仅适用于获得咖啡师技能许可的授权培训师。请不要与其他 AST 或学习者共享此文档。

Note: This guidebook replaces previous curriculum documents. This guidebook is only available to authorized trainers licensed in Barista Skills. Please do not share this document with other ASTs or learners.

1. 一般信息 General Information

<u>课程信息 Course Information</u>

课程长度：最少 21 小时 Course Length: Minimum 21 hours

*通过*以下方式花费的时间可以满足最低课程时间：*Minimum course hours* are met through time spent in the following ways:

- AST 可亲自或通过远程学习提供实时指导。The AST delivering live instruction in-person or through distance learning.
- AST 进行非正式和正式评估。（注意：这不包括学习者参加在线笔试所花费的时间。）The AST conducting informal and formal assessments. (Note: This does not include time spent by the learner taking the online written exam.)
- 学习者参与实践课程活动以强化基础知识。The learner engaging in activities that introduce and/or reinforce the theoretical or practical course objectives

报考要求 Prerequisites：咖啡师技能中级课程。咖啡入门课程，咖啡师初级课程 。建议有这些模块的所有知识和技能储备，并通过实际和/或书面评估测试。冲煮中级模块也是建议的前置课程。Barista Skills Intermediate course. Introduction to Coffee, Barista Skills Foundation are recommended. All knowledge and skills from these modules will be assumed as being held and may be tested through the practical and/or written assessments.
Brewing Intermediate is also recommended.

理想的学生应已担任咖啡师至少 12 个月，并承担一些管理职责。The ideal candidate has worked as a barista for a minimum of 12 months, in a role with some managerial responsibilities.

<u>笔试信息：Written Exam Information:</u>

在线笔试总题数：30（每题 1 分）Total Number of Questions on Online Written Exam: 30 (worth one point each)

在线笔试总时间：32 Total Time Allowed for Online Written Exam: 32 minutes

及格分数（在线笔试）：80% Passing Score (Online Written Exam): 80%

<u>实践考试信息：Practical Exam Information:</u>

实践考试总时间：95 分钟 Total Time Allowed for Practical Exam: 95 minutes

实践考试科总数：4 Total Number of Sections on Practical Exam: 4

及格分数（实践考试）：专业的咖啡师技能实践考试将测试高级咖啡认证所需的关键技能。每个部分均经过仔细校准，以评估学习者的技能能力。必须通过所有部分，才能整体通过考试。无需强求获得实践考试 100%的分数。有关部分及格分数，请参阅试卷。 Passing Score (Practical Exam): The Professional Barista Skills practical exams tests the key skills demanded of a high-level coffee professional. Each section is carefully calibrated to evaluate the learner's skill competencies. All sections must be passed in order to pass the exam as a whole. A 100% score is not expected. Please refer to the exam for section passing score

2. 课程说明和更新 Course Description and Updates

描述 Description

“高级咖啡师技能”课程旨在测试专业咖啡师（例如，具有 12 个月或以上工作经验的咖啡师）更高等级的技能和详细科学知识。成功的学生将探索并展示所预期的咖啡师高级技能。特别是，参与者将学习复杂的品尝方法和咖啡风味的描述性说明；详细了解饮料成分和可用来最大程度地提高饮料质量的技术，了解如何管理与指导其他人生产优质饮料的技能，展示出对如何制定冲煮配方的理解以及构建饮料菜单的系统方法，并最终能够始终如一地应用最高标准的拉花技术和打发牛奶技术。笔试会根据课程中的内容来测试专业知识，而实践考试则会测试课程中通过不同活动所学习的技能。 The Barista Skills Professional course is designed to test advanced skills and detailed knowledge of the science behind processes used by a professional barista (for example, a barista with 12 months or more of work experience). Successful candidates will have explored and demonstrated the advanced skills typically expected of a head barista. In particular, participants will learn a sophisticated tasting methodology and descriptive explanation of coffee flavors; gain a detailed understanding of drink ingredients and the techniques available to maximize the quality of the drinks made, understand how to manage the skills of others to produce quality drinks, demonstrate an understanding of how to develop brew recipes and a systematic method of structuring beverage menus, and finally be able to consistently apply the highest standard of latte art and milk steaming techniques. A written exam tests professional course knowledge while a practical exam tests the skills described above based on different working activities carried out during the course.

先前版本的课程更新 Curriculum Updates from Previous Version

咖啡师专业课程已经过较大的更新和修订，改进后又了更加清晰的结构。还有一些参考文献的清晰链接，供 AST 参阅。AST 应仔细阅读本节以及第 4 节中的实践课程，以确保其教材是最新的。在实践考试中已添加了详细的说明和评分，以帮助 AST 和学习者更加清晰了解考试内容。 The Barista Professional curriculum has been significantly updated and revised to provide clarity, an improved structure, and clear links to reputable references that ASTs can use to support the training they provide. ASTs should review in detail this section as well as the actual curriculum in Section 4 to ensure their teaching materials are up to date. Detailed instructions and scoring have been added to the practical exam to provide clarity to the AST and the learner.

课程补充：Additions to Curriculum:

- 附录 8-活动准备指南 Appendix 8 - Activities Preparation Guide
- 3.01 |栏目| 咖啡豆 | Section | COFFEE BEANS
    - 1 | 主题：海拔高度对咖啡风味/密度的影响 Topic: Effect of Altitude on Coffee Flavor / ensity
    - 2 | 主题：品种和育种 Topic: Variety and Cultivar
    - 3 | 主题：处理 Topic: Processing
    - 5 | 主题：包装，储存和保鲜-子主题：储存温度 Topic: Packaging, Storage and Freshness - Sub-Topic: Storage Temperatures

- 3.02 |栏目|工作空间管理和工作流程 | Section | WORKSPACE MANAGEMENT AND WORKFLOW
    - 1 | 主题：饮料生产（替换以前的 6.0 Barista Menu 中的所有目标） Topic: Drinks Production (replace all objectives from former 6.0 Barista Menu)
- 3.04 |栏目|萃取和冲煮-3 |主题：浓缩咖啡 | Section | EXTRACTION AND BREWING - 3 | Topic: Espresso Blends
- 3.05 |栏目| 感官 |主题：有机酸（以前的 1.0 咖啡豆-1.2） | Section | SENSORY| Topic: Organic Acids (former 1.0 Coffee Beans - 1.2)
- 3.07 |栏目|水质-部分与以前的 9.0 清洁，维护，故障排除和水 | Section | WATER QUALITY - section now separated from former 9.0 Cleaning, Maintenance, Troubleshooting and Water
- 3.08 |栏目|简单的财务概念-主题：成本与商品 | Section | SIMPLE FINANCIAL CONCEPTS - Topic: Cost and Goods

删除部分课程 Deletions to Curriculum

- 1.0 咖啡豆 COFFEE BEANS
    - 1.2 豆的成分及其成分对风味和醇厚度的影响 The composition of the bean and what the components contribute in flavor and body
    - 1.3 通过杯测评估不同的产地，处理和烘焙特性，以及它们如何提供不同的风味和表现 Evaluate by cupping different origins, processing and roast profiles, and how these offer different flavors and performance
- 2.0 工作空间管理 WORKSPACE MANAGEMENT
    - 分析咖啡馆布局，以确保速度和效率，良好的工作流程和流畅的客户流量。 Analyze a café layout to ensure speed and efficiency, good workflow and smooth customer traffic.
    - 找出咖啡馆布局方面的问题。 Identifies problems with café layout.
    - 调整或建议调整咖啡馆布局，以纠正与速度，效率，工作流程和客户流量有关的问题。 Adjusts or recommends adjustments to café layout to correct problems related to speed, efficiency, workflow and customer traffic.
- 3.0 研磨，装粉和填压 GRINDING, DOSING AND TAMPING
    - 3.1 - 根据与使用情况有关的优缺点选择磨豆机。 Select grinder based on advantages and disadvantages related to the needs of the situation it will be used for.
    - 3.1 - 解释不同磨豆机模型的利弊，考量它们与实际使用时的需求。可以提供有关选择磨豆机型号的见解。Explain the advantages and disadvantages of different grinder models as they relate to the needs of the situation. Can offer insight on selecting a grinder model.
    - 3.2 - 从科学的角度（电机速度，粒径，热传递，刀盘更换）的角度来看，平刀和锥刀的优点和缺点。Advantages and Disadvantages of features of flat and conical burrs from a scientific perspective (motor speed, particle size, heat dispersion, burr replacement).
    - 3.2 - 根据与将要使用的情况相关的优缺点选择刀盘的类型。可以提供有关选择磨豆机刀盘的见解。Selects type of burrs based on advantages and disadvantages related to the needs of the situation they will be used for. Can offer insight on selecting a grinder burr.

- 5.0 牛奶技术 MILK TECHNIQUES
    - 5.1 - 描述作为咖啡师如何选择各种牛奶替代品，还有这些代替品的表现情况。Describe what implications this may have for a range of milk alternatives in their performance for a barista.
- 6.0 咖啡师菜单（替换为 3.02 栏目）BARISTA MENU - (replaced for 3.02 section)
- 7.0 卫生，健康与安全-整个栏目 HYGIENE, HEALTH & SAFETY – (entire section)
- 8.0 客户服务 CUSTOMER SERVICE
- 9.0 清洁，保养，故障排除（除了水这个章节内容之外）-整个栏目 CLEANING, MAINTENANCE, TROUBLE-SHOOTING (EXCEPT WATER) - entire section
- 10.0 咖啡馆管理-整个栏目（由 3.08 代替|栏目|简单财务概念-主题：成本与商品）CAFE MANAGEMENT – (entire section replaced by 3.08 | Section | SIMPLE FINANCIAL CONCEPTS - Topic: Cost & Goods)

3. 笔试题目按主题分布 Written Exam Questions Distribution by Topic

下表列出了有关在线考试问题的关键信息。

问题池：这是每个**<u>主题</u>**中将在在线考试期间向学习者抽考的问题数量。

提出的问题：这是学习者将在在线考试中针对**<u>每个主题</u>**随机接收的问题数。这个数量由考试和模块的设计者决定，目的是确保课程的每个栏目和主题均得到适当加权。

栏目权重：每个栏目标题旁边是该栏目中的问题所代表的全部考试的百分比。在笔试或实践考试中，有些主题/目标不会被抽考到。但是培训师仍应教授这些主题/目标，因为这些信息将在以后的课程中有用。

The chart below sets forth key information regarding the online exam questions.
Question Pool: This is the number of questions **<u>per topic</u>** that are available to present to the learner during the online exam.
Questions Presented: This is the number of questions a learner will randomly receive **<u>per topic</u>** during the online exam. This number was determined by the creators for the purpose of ensuring that each section and topic of the course is weighted appropriately.
Section Weighting: Next to each section title is the percentage of the total exam represented by the questions in that section. There are some topics/objectives that are not tested during the written or practical exam. The trainer should still address these topics/objectives as the information will be useful in future courses

.考试栏目和主题 Exams Sections & Topics	问题池 Question Pool	提出的问题 Questions Presented	考试栏目和主题 Exams Sections & Topics	问题池 Question Pool	提出的问题 Questions Presented
3.01 | 栏目 | 咖啡豆 20% | SECTION | COFFEE BEANS | 20%			**3.05 | 栏目 | SENSORY3% | SECTION | SENSORY | 3%**		

1 | 主题：海拔高度对咖啡风味/密度的影响 Topic: Effect of Altitude on Coffee Flavor / Density	3	1
2 | 主题：品种和育种 Topic: Variety and Cultivar	Taught, not tested. 授课，不考。	
3 | 主题：处理 Topic: Processing	3	1
4 |主题：脱咖啡因 Topic: Decaffeination	4	2
5 |主题：包装，储藏和保鲜 Topic: Packaging, Storage and Freshness		
子主题：对排气率的影响 Sub-Topic: Effect on Degassing Rate	2	1
子主题：储存温度 Sub-Topic: Storage Temperatures	2	1
3.02| 栏目 | 工作空间管理和工作流程 | 0% | SECTION | WORKSPACE MANAGEMENT AND WORKFLOW | 0%		
1 |主题：饮料生产 Topic: Drinks Production	实践考试 Practical Exam	
3.03 | 栏目 | 浓缩咖啡 | 7% | Section | ESPRESSO PROCESS | 7%		
1 |主题：一致性分析 Topic: Consistencies Analysis		
子主题：装粉和填压 Sub-Topic: Dosing and Tamping	Taught, not tested. 授课，不考	
子主题：浪费 Sub-Topic: Waste	实践考试 Practical Exam	
2 |主题：粒径分布 Topic: Grind Particle Distribution	3	2

1 | 主题：有机酸 Topic: Organic Acids	3	1
2 |主题：感官评估 Topic: Sensory Evaluation	实践考试 Practical Exam	
3 |主题：多种咖啡的最佳平衡 Topic: Optimum Balance of Multiple Coffees	实践考试 Practical Exam	
4 | 主题：牛奶对咖啡风味的影响 Topic: Milk Effect on Coffee Flavor	授课，不考 Taught, not tested	
3.06 | 栏目 | 牛奶 | 27% | SECTION | MILK | 27%		
1 |主题：牛奶成分与处理 Topic: Milk Composition and Processing		
子主题：牛奶成分 Sub-Topic: Milk Composition	4	3
子主题：牛奶处理 Sub-Topic: Milk Processing	3	1
2 |主题：泡沫的产生，质量和稳定性 Topic: Foam Creation, Quality and Stability	2	1
3 |牛奶质量及其发泡能力 Milk Quality and Its Ability to Foam		
子主题：影响牛奶品质的因素 Sub-Topic: Factors Affecting Milk Quality	1	1
子主题：蛋白水解和脂解 Sub-Topic: Proteolysis and Lipolysis	2	1
子主题：咖啡酸度对牛奶的影响 Sub-Topic: Coffee Acidity’s Effect on Milk	Taught, not tested 授课，不考。	

3 |主题：填压和布粉工具 Topic: Tamper and Distribution Tools	活动 Activity	
3.04|栏目| 萃取和冲煮 20% | SECTION | EXTRACTION AND BREWING | 20%		
1 |主题：咖啡机 Topic: Espresso Machine		
子主题：温度和压力对冲煮的影响 Sub-Topic: Impact of Temperature and Pressure on Brewing	3	1
子主题：锅炉系统类型 Sub-Topic: Types of Boiler Systems	2	1
子主题：压力系统 Sub-Topic: Pressure Systems	3	1
2 |主题：设计冲煮配方 Topic: Designing Brew Recipes		
子主题：强度与萃取 Sub-Topic: Strength and Extraction	Practical Exam 实践考试	
子主题：选择粉量 Sub-Topic: Choosing a Dose	Practical Exam 实践考试	
子主题：选择萃取率 Sub-Topic: Choosing a Yield	Practical Exam 实践考试	
子主题：接触时间 Sub-Topic: Shot Contact Time	Practical Exam 实践考试	
3 |主题：浓缩咖啡 Topic: Espresso Blends		
子主题：拼配设计 Sub-Topic: Blends Construction	2	1
子主题：烘焙前/烘焙后 Sub-Topic: Pre/Post Roast Blending	2	1

子主题：热量对牛奶的影响 Sub-Topic: Heat's Effect on Milk	3	1
4 | SCA 泡沫标准 SCA Foam Standards	实践考试 Practical Exam	
5 | SCA 拿铁艺术标准- SCA Latte Art Standards - Free Pour	实践考试 Practical Exam	
3.07 | 栏目 | 水质 | 23% | SECTION | WATER QUALITY | 23%		
1 |主题：测量 Topic: Measurements	活动 Activity	
子主题：总溶解固体（TDS） Sub-Topic: Total Dissolved Solids (TDS)	2	1
子主题：碱度 Sub-Topic: Alkalinity	3	1
子主题：总硬度（TH） Sub-Topic: Total Hardness (TH)	4	2
子主题：pH Sub-Topic: pH	2	1
2 |主题：水过滤 Topic: Water Filtration		
子主题：水测试和过滤系统 Sub-Topic: Water Testing and Filtration systems	3	1
子主题：水过滤系统的功能和类型 Sub-Topic: Function and Types of Water Filtration Systems	3	1
3.08 |栏目|简单的财务概念 0%| SECTION | SIMPLE FINANCIAL CONCEPTS 0%		
1 |主题：成本与商品 Topic: Cost and Goods	活动 Activity	

4 |主题：萃取测量工具和技术 Topic: Extraction Measurement Tools and Techniques	3	1	**问题总数 Total Number of Questions**	67	30

4. 带有相应在线笔试问题的课程 Course Curriculum with Corresponding Online Written Exam Questions

该课程的课程设置如下，分为栏目，主题和目标。在课程的某些区域，创作者已修改了课程，以创建更合乎逻辑，更适合层次的结构。任何修订均在 *2。课程说明和更新处可以找到。*

The course curriculum is set forth below and is divided into Sections, Topics and Objectives. In some areas of the curriculum, the creators may have revised the curriculum in order to create a more logical, level-appropriate structure. Any revisions are noted in *2. Course Description and Updates.*

开发所有在线笔试题作为对特定目标的评估。这些问题已根据主题分组。主题内的所有问题均被视为主题"池"。从该库中，将随机选择一定数量的问题，并将随机分发给学习者。由于问题的随机性，如果某个特定主题的目标不止一个，则有可能无法针对该主题的所有目标测试学习者。课程解释中还包括 AST 的注释，这些注释有助于说明内容以及如何实现目标。

All online written exam questions were developed as an assessment for a specific objective. These questions have been grouped according to topic. All questions within a topic are considered the topic "pool." From this pool, a certain number of questions will be randomly selected and presented to the learner. If a particular topic has more than one objective, there is a possibility that the learner will not be tested on all objectives in the topic. This is due to the randomization of the questions from that topic. Also included in the curriculum are notes for the ASTs that help explain the content and how to achieve the objectives.

3.01 | 栏目 | 咖啡豆 COFFEE BEANS

5 个主题

主题 Topic	**子主题 Sub-Topic**	**目标 Objectives**	**AST 注意事项 AST Notes**	**在线笔试题 Online Written Exam Questions**	**参考资料 References**
3.01.01 **海拔高度对咖啡风味/密度的影响** **Effect of Altitude on Coffee Flavor / Density**		1 了解海拔高度对咖啡的酸度，甜度和密度的影响。 Understand general effect of altitude on perception of acidity, sweetness and density of coffee	海拔升高的影响会影响咖啡樱桃的成熟率。预期的结果也会对 所得咖啡樱桃的酸度和甜度产生影响。以及对平均种子（豆）密度的预期影响。The effect of increased altitude influences the rate of cherry maturation. There is also an expected effect on acidity and sweetness in the resulting cherry. and an expected effect on average seed (bean) density.	**问题 100008881309** 阿拉比卡咖啡在海拔 1800 m 的生长情况与在海拔 1200 m 相比如何? **在 1800 m 生长的咖啡可能比在 1200 m 上生长的具有更复杂的酸度。** 在 1800 m 生长的咖啡咖啡因含量可能较高。 在 1800 m 生长的咖啡密度可能低于在 1200 m 生长的植物。 海拔高度不会导致咖啡豆的质量差异。 What is the difference between coffee beans grown at 1800 m.a.s.l. and 1200 m.a.s.l. (meters above sea level)? **A coffee plant grown at 1800 m.a.s.l. is likely to have more complex acidity than the plant grown at 1200 m.a.s.l.** A coffee plant grown at 1800 m.a.s.l. is likely to have a higher caffeine content. A coffee plant grown at 1800 m.a.s.l. is likely to have lower density than the plant grown at 1200 m.a.s.l. Altitude does not contribute to quality difference in coffee beans.	Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 21-23 Illy & Viani, Espresso Coffee, 34-35 Roast Magazine, The Book of Roast, 52

				问题 ID 100008881310 在高海拔地区种植的咖啡植物中，影响咖啡豆品质发展（例如酸度和果味）的最重要因素是什么？ 土壤 **平均温度** 湿度 遮荫 Which is the most important factor influencing the bean development in a coffee plant grown at high altitude, leading to positive quality attributes such as acidity and fruitiness? Soil **Average temperature** Humidity Shade	
				问题 ID 100008881311 海拔高度对咖啡质量有何影响？ 海拔高度不会导致咖啡豆的质量差异。 **海拔越高，咖啡的甜度越高，酸度也越高。** 较高的高度会增加咖啡中的咖啡因含量。 更高的高度会缩小咖啡豆的尺寸，使咖啡中的风味更加浓郁。 What is the effect of altitude on the quality of coffee? Altitude does not contribute to quality difference in coffee beans. **Higher altitude will contribute to higher sweetness and more complex acidity in coffee.** Higher altitude increases the caffeine content in coffee. Higher altitude will shrink the bean size, making the flavors in coffee more concentrated.	

3.01.02 **品种和育种** **Variety and Cultivar**		1. 了解品种和育种之间的区别 Understand the difference between a variety and cultivar 2. 与实验室影响相比，了解突变的概念 Understand concept of mutation as compared to lab influence	品种是： A variety is: • 一致：可以通过一组特征精确地描述，并且所有此类植物看起来都一样。 • 区别：根据上述特征，可与其他品种区分开。 • 稳定：可以以其特性对于下一代不变的方式进行复制。 • Uniform: it can be precisely described by a set of characteristics and all the plants of this type look the same. • Different: it is distinguishable from other varieties based on the above set of characteristics. • Stable: it can be reproduced in such a manner that its characteristics are unchanged to the next generation. 一个育种是： • 通过园艺或农业技术生产的任何品种，通常不在自然种群中发现；栽培品种。 • 我们在特种咖啡中知道的大多数品种都是栽培品种。 A cultivar is: • Any variety produced by horticultural or agricultural techniques and not normally	**授课，不考。Taught, not tested.**	Hoffmann, J. The world atlas of coffee, 22 SCA, The Coffee Biology Field Glossary, 51 https://worldcofferesearch.org/work/next-generation-f1-hybrids/f1-hybrids-explainer/ Roast Magazine, The Book of Roast, 44, 47-48, 53

			found in natural populations; a cultivated variety. • Most of the varieties we know in specialty coffee are cultivars.		
3.01.03 **处理** **Processing**		1. 了解特定咖啡处理方法和技术的一般术语和概念 Understand general terminology and concepts of specific coffee processing methods and techniques	发酵 • 发酵是细菌，酵母或其他微生物对物质的化学分解，通常涉及发泡反应和放热。 • 有氧-富氧环境 • 厌氧–环境中无氧 • 碳–富含二氧化碳的环境 Fermentation • Fermentation is the chemical breakdown of a substance by bacteria, yeasts, or other microorganisms, typically involving effervescence and the giving off of heat • Aerobic - oxygen rich environment • Anaerobic – absence of oxygen in environment • Carbonic – carbon dioxide rich environment 浸渍 • 浸渍是比发酵更具指向性的术语，是指微生物的新陈代谢-整个水果未经处理完整发酵。	**问题 ID 100008881312** 选择以下最能描述水洗咖啡的厌氧发酵过程的陈述。 厌氧发酵比有氧发酵更快。 去皮后，将咖啡放在敞开的容器中。 **在此过程中缺少氧气可能会大大改变咖啡的风味。** 它对风味的发展没有影响。 Select the statement below that best describes the anaerobic fermentation process of washed coffee. Anaerobic fermentation is faster than aerobic fermentation. The coffee is placed in an open tank after pulping. **The lack of oxygen in this process is likely to significantly alter the flavor profile of the coffee.** It has no impact on flavor development. **问题 ID 100008881313** 什么是厌氧发酵? 咖啡在充满水的发酵罐中处理。 咖啡与果皮一起发酵。 在阳光下将咖啡发酵，直到咖啡樱桃完全干燥。 **咖啡在完全密封且缺氧的发酵罐中处理。** What is anaerobic fermentation? Coffee is processed in a water filled fermentation tank. Coffee is fermented together with the pulp of the cherries. Coffee is fermented under the sun until the cherries dry completely. **Coffee is processed in a fully sealed and oxygen deprived fermentation tank.**	https://dailycoffeenews.com/2019/06/03/a-guide-to-carbonic-maceration-and-anaerobic-fermentation-in-coffee/ https://www.ptscoffee.com/blogs/news/experimental-coffee-processing-methods-explained

			Maceration • Maceration is a more inclusive term than fermentation, referring to microbial metabolism - whole fruit being fermented without processing	**问题 ID 100008881314** 以下陈述是对还是错？乳酸发酵允许乳酸菌在厌氧条件下生长。 **对** 错 Is the following statement true or false? Lactic fermentation allows for the growth of lactic acid bacteria under anaerobic conditions. **True** False	
		2. 回顾不同的混合处理过程及其对咖啡的影响 Recall different hybrid processes and their effect on the coffee	• 去皮日晒 Pulped Natural • 蜜处理-和颜色定义（国家/地区差异） Honey Process – and color definitions (country variations) • 半水洗/湿刨 Semi-washed / wet-hulled	**授课，不考。** **Taught, not tested.**	Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 51-70 Hoffmann, J. The world atlas of coffee, 31-37 Illy & Viani, Espresso Coffee, 91-100
3.01.04 **脱咖啡因** **Decaffeination**		1. 了解咖啡生豆的脱咖啡因过程，特别是基于溶剂和非溶剂的过程 Understand the decaffeination process of green coffee specifically solvent and non-solvent based processes	• 对比与未调整过的咖啡相比，去咖啡因的咖啡中通常含有多少咖啡 • 脱咖啡因作用对萃取和风味的影响 溶剂法 • Contrast how much caffeine typically is in a decaffeinated coffee as compared to coffee that has been unaltered. • Include effects of decaffeination on extraction and flavor	**问题 ID 100008881328** 下列哪种溶剂**不**用于脱咖啡因咖啡？ 二氯甲烷 **乙酸** CO_2 瑞士水工艺 Which solvent below is **NOT** used to decaffeinate coffee? Dichloromethane **Acetic acid** CO_2 Swiss Water process	Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 225-241 Stephenson, T., The curious barista's guide to coffee, 39-40 Illy & Viani, Espresso Coffee, 143-149 Christine C, Barista Bible, 17

			• 溶剂工艺 Solvent based process ○ 用化学溶剂从咖啡豆中除去咖啡因；即二氯甲烷，乙酸乙酯 Caffeine is removed from the beans with a chemical solvent; i.e. methylene chloride, ethyl acetate ○ 通过直接或间接联系完成 Done by direct or indirect contact • 非溶剂工艺： Non-solvent based process: ○ 瑞士水 Swiss Water ○ 二氧化碳 Carbon dioxide	**问题 ID 100008881330** 以下陈述是对还是错？以水处理法去除咖啡因的咖啡会损失可溶性固体，从而导致醇厚度变薄。 Is the following statement true or false? Water-decaffeinated coffee can suffer from a loss of soluble solids leading to a thin body. **对 True** 错 False	Roast Magazine, The Book of Roast, 371-377 Clarke, R. and Vitzthum, R. Coffee Recent Developments, page 108-118
				问题 ID 100008881334 以下哪种溶剂通常**不**用于脱咖啡因作用？ CO_2 乙酸乙酯 **液氮** 水 Which of the following solvents is **NOT** commonly used for decaffeination? CO_2 Ethyl acetate **Liquid nitrogen** Water	

				问题 100008881335 章 以下哪项被称为“有机认证友好型脱咖啡因法”? 苯乙酸乙酯 乙酸乙酯 **CO_2** 甘油三酯 Which of the following is known as an Organic Certification Friendly decaffeination method? Benzene Ethyl acetate **CO_2** Triglyceride	
3.01.05 **包装，储藏和新鲜** **Packaging, Storage and Freshness**	**3.01.05.01** **对排气率的影响** **Effect on Degassing Rate**	1. 描述不同的包装和技术如何影响烘焙咖啡豆的排气速率 Describe how different packing and technology affects the rate of degassing of roasted coffee beans	包装注意事项: ● 单向阀-允许 CO2 逸出，但阻止氧气进入。 ● 真空密封罐 ● 牛皮纸袋与铝箔纸相比 ● 可降解袋 ● 充氮气 Packaging considerations: ● One-way valves - allows CO2 to escape but keeps oxygen out. ● Vacuum sealed cans ● Kraft brown bags compared to foil ● Compostable bags ● Nitrogen flushing	**问题 ID 100008881337** 充氮气（包装咖啡时用氮气代替空气）如何影响咖啡的保质期? **保质期一般较长** 保质期一般较短 保质期不受影响 How does nitrogen flushing (replacing air with nitrogen when packing coffee) impact the shelf life of coffee? **Shelf life is generally longer** Shelf life is generally shorter Shelf life is not impacted	Christine C, Barista Bible, 29 Illy & Viani, Espresso Coffee, 245-255 SCA, The Coffee Freshness Handbook, 19-20, 48 Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 340-350

				问题 ID 100008881338 为了确保更长的保鲜期，通常使用以下哪两种包装类型? 牛皮纸袋 **充氮气包装** **真空包装** 夹链袋 Which **two (2)** of the following packaging types are usually used to ensure longer duration of freshness? Kraft paper bag **Nitrogen flushed packaging** **Vacuum packaging** Zip lock bag	Roast Magazine, The Book of Roast, 379-385
		2. 通过萃取过程中的视觉线索和风味来区分过新鲜或陈旧的咖啡 Distinguish coffee that is too fresh or stale, by visual clues during its extraction and by its flavor	AST 讨论： ASTs to discuss: • 在任何给定的烘焙日期中，烘焙咖啡中残留的二氧化碳量如何影响浓缩咖啡的萃取/流速 • 如何调整冲煮配方以解决此问题 • How the amount of CO_2 that is retained within roasted coffee on any given roast date affects espresso extraction/ flow rate • How adjustments to brew recipes are necessary to account for this	**授课，不考。Taught, not tested.**	

	3.01.05.02 **储存温度** **Storage Temperatures**	1. 了解低温保存烘焙咖啡的影响 Understand the impact of storing roasted coffee at low temperature		**问题 100008881341** 将咖啡豆存储在冰箱中时，以下哪项是重要的考虑因素? 使用真空密封容器 仅在冲煮时才将豆从冷冻室中取出 冷冻咖啡豆的量 **以上所有** Which of the following are important considerations when storing coffee beans in the freezer? Use of vacuum sealed container Removal of beans from freezer only when they are to be brewed Size of portion to be frozen **All of above**	Rao, The Professional Barista's Handbook, 78 Fury, M., *To Freeze Or Not To Freeze, That Is The Coffee Question* Dockrill, P., *Want To Drink Better-Tasting Coffee? Freeze Your Beans, Say Scientists* Huladaddy.com. *Freezing Coffee Beans Keeps Them Fresh* Compound Coffee, *Does Grinding Frozen Coffee Beans Impact Brewing & Taste?* Sivetz, M. and Desrosier, *Coffee Technology*, 280
				问题 ID 100008881342 以下陈述是对还是错? 冷冻烘焙好的咖啡可以大大减少挥发性化合物的移动，因此可以作为长期储存方法。 **对** **错** Is the following statement true or false? Freezing roasted coffee greatly reduces the movement of volatile compounds, therefore it can work as a long-term storage method. **True** False	

3.02 |栏目|工作空间管理和工作流程 WORKSPACE MANAGEMENT AND WORKFLOW

1 主题

主题 Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考资料 References
3.02.01 **饮料生产** **Drinks Production**	1. 根据 SCA **饮料标准**一致、高效地生产**数杯浓缩咖啡** Produce **multiple espressos** consistently and efficiently according to the SCA Drink standard 2. 根据 SCA **饮料标准**一致，高效地生产**多种牛奶饮料** Produce **multiple milk beverages** consistently and efficiently according to the SCA Drink standard 3.根据 SCA **饮料标准**快速有效地准备复杂的饮料订单 Apply techniques to prepare **complex orders of drinks** quickly and efficiently according to the SCA Drink standard	• 期望这一级别的学习者以系统且可重复的方式创建和优化意式浓缩咖啡配方，但不一定受固定范围或技术的限制。 • 在咖啡师技能高级实践考试中可以找到允许的粉量，溢出量和接触时间的差异。 • 有关饮料规格，请参见 SCA 咖啡师饮料标准。 • 有关温度，填充量和废料的允许差异，可在咖啡师技能专业实践考试中找到。 • Learners at this level are expected to create and optimize espresso recipes in a systematic and repeatable way but are not necessarily constrained by fixed ranges or techniques. • The tolerances allowed for dose, yield and contact time can be found in the Barista Skills Professional Practical Exam. • See SCA Barista Drink Standards for drinks specification. • The tolerances allowed for temperature, fill level and waste can be found in the Barista Skills Professional Practical Exam.	**在实践考试中考核。** **Tested on Practical Exam.**	Barista Skills Professional Practical Exam (Trainers version) SCA Drink Standards

3.03 |栏目|浓缩咖啡流程 ESPRESSO PROCESS

3 个主题

主题 Topic	**子主题** Sub-Topic	**目标** Objectives	**AST 注意事项** AST Notes	**在线笔试题** Online Written Exam Questions	**参考资料** References
3.03.01 **一致性分析** **Consistencies Analysis**	**3.03.01.01** **装粉和填压** **Dosing and Tamping**	1．认识到其他咖啡师调制的饮料在装粉和填压中存在不一致之处，以及对萃取的影响. Recognize inconsistencies in dosing and tamping in drinks prepared by other baristas and the impact on extraction	着重于学习者的发展能力，以评估两组萃取之间的装粉和布粉技术的差异，确保两组萃取时间误差不超过三秒。 Focus on learner developing ability to evaluate differences in dosing and distribution techniques within pairs, and track pairs of espresso extraction within 3 seconds.	**授课，不考。Taught, not tested.**	

	3.03.01.02 **浪费** **Waste**	1. 识别浓缩咖啡生产过程中的废料来源和废料量 Recognize sources of waste and waste amounts during the process of producing espressos	Coffee waste can be found across the entire coffee station: 在整个工作区域内都可以发现咖啡残渣: -咖啡机 -磨豆机粉仓室 -敲渣区 - 在操作台上 -在垃圾桶里 - 在地上 -布粉工具 -或其他地方 - espresso machine - grinder dosing chamber - knock box area - on the counter - in the trash - on the floor - distribution tools - or elsewhere 在 0.0 g 到 5.0 g 的残粉区间评估分别制作 4 杯饮料的残粉程度，以 0.5g 为一个量级（AST 应注意，该概念将在实际考试中评估）。 Evaluate 0.5 g increment differences of waste between 0.0 g to 5.0 g per 4 beverages made (AST should note this concept will be assessed on Practical Exam).	**在实践考试中考核。Tested on Practical Exam.**	Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 321-322

3.03.02 **粒径分布** **Grind Particle Distribution**		1.认识到磨豆机变量和烘焙水平对研磨颗粒分布的影响 Recognize impact of grinder variables and roast level on grind particle distribution	以下变量会影响粒径分布: The following variables can influence particle size distribution: 1. 磨豆机电机转速 Motor speed of grinder 2. 刀盘设计 Burr design 3. 豆温度 Bean temperature 4. 烘焙度 Roast degree	**问题 ID 100008892600** 以下陈述是对还是错? 与较深的烘焙咖啡相比， 研磨较浅的烘焙咖啡需要更多的能量，并且粒径分布范围更窄（假设所有其他变量保持不变）。 **对** 错 Is the following statement true or false? Compared with darker roasted coffees, grinding lighter roasted coffees requires more energy, and results in a narrower range of particle size distribution (assuming all other variables remain the same). **True** False	Rao, The Professional Barista’s Handbook, 6-9 Hoffmann, J. The world atlas of coffee, 68-69 Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 315-325 Christian Klatt, Barista Camp 2015, *Heating (In) Grinders*
				问题 ID 100008892604 在以下选项中，这些微米尺寸的研磨咖啡范围中哪一个最适合用于浓缩咖啡萃取? Of the following options, which of these ground coffee size ranges is the most suitable for espresso extraction? 950-1150 微米 microns 650-850 微米 microns **250-450 微米** microns 750-950 微米 microns	

				问题 ID 100008892605 下列哪种说法是正确的? 咖啡熟豆的溶解度和感官品质仅受烘焙颜色的影响。 **研磨过程中的粒径分布受烘焙程度的影响。** 烘焙方法会影响浓缩咖啡的感官品质，但不会影响溶解度。 Which of the following statements is correct? Solubility and sensory qualities of roasted coffee are only affected by roast color. **Particle size distribution during grinding is affected by roast degree.** The roasting methodology will affect the sensory qualities of espresso but will not affect solubility.	
3.03.03 **填压和布粉工具** **Tamper and Distribution Tools**		1 意识到布粉工具和不同的填压设计，以及它们对咖啡粉饼和咖啡师日常工作效率的影响. Aware of distribution tools and different tamper designs, and their influence on the coffee bed and barista routine efficiency	• 一致性可能会增加，而效率可能会下降。 • 评估布粉工具对萃取的影响 • 不同样式的压粉器的功能 • Consistency can potentially increase while efficiency can decrease. • Evaluate the cause and effect of distribution tools on a dose • Features of different styles of hand tampers	**该主题有一个活动。** **This topic has an activity.** **见 8。附录-活动准备指南** **See 8. Appendix - Activity Preparation Guide**	Coffee for us, *10 Best Espresso Tampers In 2020 Reviewed* Socratic Coffee, *Comparing the Impact Of Tamper On Extraction*

3.04 | 栏目|萃取和冲煮 EXTRACTION AND BREWING

4 个主题

主题 Topic	子主题 Sub-Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考资料 References
3.04.01 **咖啡机** **Espresso Machines**	**3.04.01.01** **温度和压力对冲煮的影响** **Impact of Temperature and Pressure on Brewing**	1. 认识到温度对萃取和流速的影响 Recognize the influence of temperature on extraction and flow rate	• **压力**调节可能会导致流速和化合物溶解度增加或减少。 • **温度**调节可能会导致流速和化合物溶解度增加或减少。 • 质量与能量之比的变化（粉量与热水温度之间的关系）将改变萃取率。 • Adjustments in **pressure** may cause the flow rate and dissolving of compounds to increase or decrease. • Adjustments in **temperature** may cause the flow rate and dissolving of compounds to increase or decrease. • Changes to the mass to energy ratio (the relationship between the dose and the temperature of the hot water) will change extraction %.	**问题 ID 100008892610** 选择最能表达事实的答案。在较高温度下： 水的动能较低， 水的溶解度下降， 水的粘度较高， **浓缩咖啡萃取率会增加。** Select the answer that best completes the statement. At higher temperatures _________ _________ the kinetic energy of the water is lower. water's solvency decreases. water's viscosity is higher. **the extraction rate in an espresso brew will likely increase.** **问题 ID 100008892614** 降低冲煮温度将如何影响浓缩咖啡的萃取率？ **萃取率可能会降低，** 萃取率可能会增加， 萃取率不会受到影响， 萃取率会根据烘焙曲线而波动。 How will decreasing the brewing temperature impact the extraction rate of espresso? **Extraction rate will likely decrease.** Extraction rate will likely increase. Extraction rate will not be affected. Extraction rate will fluctuate according to the roast profile.	Illy & Viani, Espresso Coffee, 260-270, 274-279, 282-286 Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 365-375 Stephenson, T., The curious barista's guide to coffee, 108

		2. 认识到压力对萃取和流速的影响 Recognize the influence of pressure on extraction and flow rate		**问题 ID 100008892619** 增加水压将如何影响萃取率？ 萃取率可能会降低， **萃取率可能会增加，** 萃取率不会受到影响， 萃取率会根据烘焙曲线而波动。 How will increasing water pressure impact the extraction rate? Extraction rate will likely decrease. **Extraction rate will likely increase.** Extraction rate will not be affected. Extraction rate will fluctuate according to the roast profile.	
	3.04.01.02 **锅炉系统类型** **Types of Boiler Systems**	1. 认识到以下各项之间的基本区别及其对温度稳定性的影响： • 单锅炉 • 多锅炉 • P.I.D.系统 Recognize basic differences between: • Single boiler • Multiple boiler • P.I.D.systems and their influence on temperature stability		**问题 ID 100008892654** 下图所示的锅炉系统的主要优点是什么？ **水的热稳定性** 减少单个锅炉元件上的负载 将电气负载分散到整个机器上 降低功耗 What is the main benefit of the type of boiler system shown in the picture below?  **Thermal stability of the water** Reduce load on individual boiler elements Spread electrical load across the machine Reduce power consumption	Rao, *The Professional Barista's Handbook*, 21

				问题 ID 100008892656 选择最能完成以下陈述的正确答案: ________提供的浓缩咖啡机中水温控制的方法不太复杂，也不太精确。 **压力或恒温** 器数字温度计 PID 控制器 模拟温度计 Select the correct answer that best completes the following statement: A (An)_______ provides a less sophisticated, and less precise method of water temperature control in an espresso machine. **Pressure stat or thermostat** Digital thermometer PID Controller Analog thermometer	
	3.04.01.03 **压力系统** **Pressure Systems**	1. 认识到压力和浸泡的基本区别和机械应用 • 逐渐增加压力 • 初始管路压力 • 一致的压力系统 Recognize basic differences and mechanical application of pressure and infusion on	不同的机器将以不同的方式施加和控制管路还有泵压力。压力来源: 冲煮用水的两种主要压力来源：管路压力和泵压力。 描述如何进行预浸泡（通过杠杆系统或电动泵系统）。	**问题 ID 100008892657** “管线压力”指的是什么? 浓缩咖啡机泵施加的压力。 **来自当地供水的压力。** 水平放置进气软管时产生的压力。 手动机械手柄向上运动时产生的压力。 What does “line pressure” refer to? The pressure exerted by the espresso machine pump. **The pressure originating from the local water supply.** The pressure created when placing the inlet hose horizontally. The pressure created during the upward motion of a manual machine lever.	Illy & Viani, Espresso Coffee, 283-284

		• Gradual ramp up pressure • Initial line-pressure and • Consistent pressure systems	描述冲煮头工作时一致压力。 压力和流量的应用 高压与低压冲煮-流量/萃取 某些机器可能有在整个意式浓缩咖啡萃取过程中调整压力的功能。 Different machines will apply and control line and pump pressure in different ways. **Sources of pressure:** The 2 main sources of pressure applied to water for brewing: Line pressure and Pump pressure. Describe how pre-infusion can happen (via lever system or motor pump system). Describe consistent pressure when the group is engaged. **Application of pressure and flow** Brewing with higher vs. lower pressure – impact on flow rate/ extraction Manipulating pressure throughout the espresso shot can happen with some machines.	**问题 ID 100008892658** 萃取浓缩咖啡时使用管路压力的预期好处是什么？ 它降低了功耗。 它增加了浓缩咖啡机的使用寿命。 **它可以使咖啡粉饼轻柔地润湿。** 它降低了泵的噪音。 What is the intended benefit of using line pressure when extracting espresso? It reduces power consumption. It increases the usable life of the espresso machine. **It allows for gentle wetting of the coffee cake.** It reduces pump noise. **问题 ID 100008892659** 选择最能使句子完整的答案。浓缩咖啡机通常使用 _______________将水推入并通过咖啡粉饼。 重力压力 热压力 真空压力 **管路和/或泵压力** Select the answer that best completes the sentence. An espresso machine typically uses _______________ to push water into and through the coffee cake. Gravity pressure Thermal pressure Vacuum pressure **Line and/or pump pressure**	

3.04.02 设计冲煮配方 **Designing Brew Recipes**	**3.04.02.01 浓度与萃取** **Strength and Extraction**	1. 校准多种咖啡的冲煮配方，并跟踪其浓度和萃取百分比 Calibrate different brew recipes for multiple coffees and track the strength and extraction percentages		**在实践考试中考核。** **Tested on Practical Exam.**	
	3.04.02.02 选择粉量 **Choosing a Dose**	1. 选择适合特定咖啡的粉量 Choose the appropriate dose for a specific coffee	• 设计浓咖啡配方时，要从粉量开始。 • 粉量取决于粉碗的大小。 • 粉量会影响冲煮温度和流速。 • 当使用较低粉量时，需要更细的研磨才能达到一定的喷射时间-可能导致更高的萃取率 • Designing an espresso recipe, starts with the dose amounts. • Dose amounts are dependent on basket size . • Dose influences brew temperature and flow rate. • When using lower doses, a finer grind is needed to achieve a certain shot time - potentially leading to a higher extraction.	**在实践考试中考核。** **Tested on Practical Exam.**	Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 326

	3.04.02.03 **选择萃取率** **Choosing a Yield**	1. 确定特定咖啡的合适冲煮率/ EBF Identify the appropriate Brew Ratio/EBF for a specific coffee	• 不同的冲煮比例（使用相同粉量）将影响浓缩咖啡的萃取和浓度（风味/质地）。 • 选择萃取率需要了解咖啡的产地，过程和烘焙程度。 • Different brew ratios (using the same dose) will influence extraction and strength of an espresso (flavor/ texture) . • Choosing a yield involves an understanding of coffee origin, process, roast degree.	**在实践考试中考核。Tested on Practical Exam.**	
	3.04.02.04 **萃取接触时间（秒）** **Shot Contact Time (seconds)**	1. 了解长和短浓咖啡接触时间对特定咖啡的风味/质地的影响 Understand the effect of long and short espresso shot contact time on flavor/texture for a specific coffee	• 不同的接触时间（使用相同的粉量和萃取率）会影响咖啡的萃取和强度（风味/质地）。 Different shot contact time (using the same dose and yield) can influence extraction and strength of an espresso (flavor/ texture).	**在实践考试中考核。Tested on Practical Exam.**	Illy & Viani, Espresso Coffee, 285

3.04.03 **浓缩咖啡** **Espresso Blends**	**3.04.03.01** **拼配设计** **Blends Construction**	1. 了解浓缩咖啡拼配的构成方式 Understand how espresso blends are constructed	利用咖啡生豆/烘焙水平/咖啡萃取的所有信息，介绍如何创建浓缩咖啡拼配的不同方法。 说明制作拼配配方时使用的一些技术： • 杯测 • 小规模混合 Utilizing all information of green coffee /roast level /extraction of coffee, introduce different approaches on how to create an espresso blend. Explain some of the techniques used when creating a coffee blend: • Cupping • Small scale blending	**问题 ID 100008892660** 当其中一种成分的比例不足总比例的 5%时，会对咖啡拼配产生什么影响？ 拼配的生产可能很复杂， **每次萃取中的意式浓缩咖啡不太可能含有相同的 5%，** 浓缩咖啡的萃取将非常一致。 浓缩咖啡的味道会更加复杂。 What could be the impact on a coffee blend when one of the components represent less than 5% of the overall blend? The blend can be complex to produce. **The espresso is unlikely to contain the same 5% in each extraction.** The espresso extraction will be very consistent. The espresso taste will be more complex.	Illy & Viani, Espresso Coffee, 141-142 Roast Magazine, The Book of Roast, 357-358 Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 268
				问题 ID 100008892661 萃取由多种比例的多种咖啡制成的拼配时，需要注意什么？ **每份浓缩咖啡的萃取可能会不一致** 拼配的生产可能很复杂， 拼配的生产成本可能很高， 复杂拼配没有不利之处。 What is a concern when extracting a blend made of multiple coffee components at different ratios? **There may be inconsistency from one espresso extraction to another.** The blends may be complex to produce. The blend may be costly to make. There is no downsides to complex blends.	

	3.04.03.02 **烘焙前/烘焙后** **Pre / Post Roast Blending**	1. 了解烘焙前/烘焙后混合的区别 Understand the differences between pre/post roast blending	熟拼提供了最佳烘焙每个成分的机会，但具有额外的生产步骤的成本。 咖啡品类的数量会影响混合： • 当拼配的百分比变得太小时，在每个粉碗中获得正确的百分比的可能性就很小。 Post blending provides the opportunity to roast each component optimally but has the cost of extra production steps. The number of components impact a blend: • When % of a blend becomes too small, the likeliness of obtaining the right % in every portafilter or brew-basket is small.	**问题 ID 100008892662** 以下陈述是对还是错？ 熟拼提供了最佳烘焙每个成分的机会，但具有额外的生产步骤的成本。 Is the following statement true or false? Post blending provides the opportunity to roast each component optimally but has the cost of extra production steps. **True** **对** False 错	
				问题 ID 100009359968 一位烘焙师希望用两种单一产地咖啡豆制成拼配，必须决定要生拼还是熟拼。如果选用熟拼的话，需要考虑哪两个因素？ 两个咖啡有不同的杯测结果 两个咖啡来自不同农场 **两个咖啡有不同的密度** **两个咖啡有不同的含水率** A roaster wishes to construct a blend with 2 single origin coffees, and they have to choose between pre-roast blending and post roast blending. Which two (2) factors should be considered if post-roast blending is chosen? That both coffees have different cupping scores. That both coffees are from different farms. **That both coffees are of different densities.** **That both coffees have different moisture content.**	

3.04.04 **萃取测量工具和技术** **Extraction Measurement Tools and Techniques**		1. 利用萃取测量工具和技术（浓缩咖啡冲煮图和公式，测量设备和软件）来评估浓缩咖啡 Utilize extraction measurement tools and techniques (espresso brewing charts and formulas, measurement devices and software) to evaluate espresso	AST 演示使用此类工具的测量过程 AST to demonstrate process of measurement using such tools	**问题 ID 100008892664** 在折光仪上测量样品之前，必须做的一件事是什么？ 摇晃杯子以分离出油脂。 趁热测量样品，以保存尽可能多的油脂。 **将咖啡冷却至室温.** 等待 24 小时再进行测量。 What is the one thing necessary to do prior to testing a sample on a refractometer? Swirl the shot to separate the crema. Test the sample while hot to capture as many oils as possible. **Cool the shot to room temperature.** Wait 24 hours to test the shot.	
				问题 ID 100008892665 为什么在使用折光仪进行测量之前，必须先将义式浓缩咖啡进行过滤？选出所有正确的答案。 冷却下来 **去除气体含量** **去除不溶性固体** **去除油脂** Why might espresso brewed coffees be filtered prior to being tested with a refractometer? Select **ALL** that are correct. To cool it down **To remove gas content** **To remove insoluble solids** **To remove oils**	

				问题 ID 100008892666 什么原因会导致折光仪测量结果不一致? 电池电量过低 校准不正确 取样处理不一致 样品温度过高 **以上皆是** What could cause refractometer inconsistencies? A low battery Out of calibration Inconsistent handling High sample temperature **All of above**	

3.05 | 栏目|感官 | SENSORY

4 主题

主题 Topic	**目标** Objectives	**AST 注意事项 AST** Notes	**在线笔试题** Online Written Exam Questions	**参考资料** References
3.05.01 **有机酸** **Organic Acids**	列出咖啡中最常见的主要有机酸。即柠檬酸，苹果酸，醋酸，酒石酸 List primary organic acids most commonly found in coffee. i.e. Citric, Malic, Acetic, Tartaric		**问题 ID 100008892667** 将以下有机酸与其相应的典型味道相匹配: a）柠檬酸 b）苹果酸 c）乙酸 d）绿原酸 1）青苹果 2）苦/涩 3）醋 4）柠檬 **A = 4 / B = 1 / C = 3 / D = 2** Match the following organic acids with their corresponding typical taste: a) Citric b) Malic c) Acetic d) Chlorogenic 1) Green apples 2) Bitter/astringent 3)Vinegar 4) Lemon	Illy & Viani, Espresso Coffee, 54, 152-153
			问题 ID 100008892668 咖啡中通常发现哪些有机酸? 水杨酸，柠檬酸和硼酸 奎宁酸，乙酸和硝酸 磷酸，苹果酸和盐酸 **柠檬酸，苹果酸和酒石酸** Which organic acids are commonly found in coffee? Salicylic acid, citric acid and boric acid Quinic acid, acetic acid and nitric acid Phosphoric acid, malic acid and hydrochloric acid **Citric acid, malic acid and tartaric acid**	

			问题 ID 100008892669 咖啡中不太可能发现哪种酸? 苹果酸 酒石酸 乳酸 **硼酸** Which acid is not likely to be found in coffee? Malic acid Tartaric acid Lactic acid **Boric acid**	
3.05.02 **感官评估** **Sensory Evaluation**	1. 使用三角杯测法确定一种咖啡是否比另一种咖啡更酸，更甜，更苦和/或更陈旧 Identify if a coffee is more sour, sweet, bitter, and/or stale than another coffee using triangulation		**在实践考试中考核。** **Tested on Practical Exam.**	
3.05.03 **多种咖啡的最佳平衡** **Optimum Balance of Multiple Coffees**	1. 确定特定咖啡的酸度，甜度和苦味的最佳平衡 Identify optimum balance of acidity, sweetness, and bitterness for a specific coffee		**在实践考试中考核。** **Tested on Practical Exam.**	Illy & Viani, Espresso Coffee, 265, 339-343
3.05.04 **牛奶对咖啡风味的影响** **Milk's Effect on Coffee Flavor**	1. 识别使用不同类型的奶（牛奶和牛奶替代品）和不同的牛奶与浓缩咖啡比例所带来的不同风味结果 Identify different flavor outcomes of different coffees with different types of milk (cows and milk alternatives) and different milk to espresso ratios		**授课，不考。Taught, not tested.**	Illy & Viani, Espresso Coffee, 311-313

3.06 | 栏目|牛奶 MILK

5 个主题

主题 Topic	子主题 Sub-Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考资料 References
3.06.01 **牛奶成分与处理** **Milk Composition and Processing**	**3.06.01.01** **牛奶成分** **Milk Composition**	1. 描述牛奶中脂肪的性质-大小，组成 Describe the nature of fat in milk: size, composition	牛奶中的脂肪存在于每个被脂肪膜包裹的小球中。 脂肪以甘油三酸酯的形式存在，是由游离脂肪酸和甘油组成的分子。 当脂肪开始分解时，游离脂肪酸会抑制泡沫。 Fat in milk exists in globules each encased by a fat membrane. Fat exists as triglyceride, molecules made up of free fatty acids and glycerol. Free fatty acids inhibit foaming when the fat starts to break down.	**问题 ID 100008892670** 以下陈述是对还是错？牛奶中的脂肪小球会因冷冻而受损，但能够耐受高温。 **对** **错** Is the following statement true or false? Fat globules in milk are damaged by freezing but are able to tolerate high temperatures. **True** False	https://scanews.coffee/2014/09/15/milk-foam-creating-texture-and-stability/ https://scanews.coffee/2014/09/15/milk-foam-creating-texture-and-stability/ Harold McGee, Food & Cooking an Encyclopedia of Kitchen Science, History and Culture, 18
				问题 ID 100008892671 乳脂中的甘油三酸酯分子由以下哪个组成： **游离脂肪和甘油** 酪蛋白和乳清 氨基酸 酵素 Triglyceride molecules in milk fat are made up of which of the following: **Free fatty and glycerol** Casein and whey Amino Acids Enzymes	

		2. 了解牛奶中的关键蛋白质 Understand the key proteins in milk	注意酪蛋白和乳清蛋白的功能和行为。 不同品种牛的蛋白质水平可能有所不同，但通常约为 3-4%。 不同的植物性奶天然具有不同的蛋白质百分比，但通常会添加其他蛋白质以辅助起泡。 Be aware of the function and behavior of casein and whey protein. Protein levels in different breeds of cow may vary but are typically around 3-4%. Different vegetable-based milks have different protein percentages naturally, but often additional proteins are added in to aid foaming.	**问题 ID 100008892672** 蛋白质占全脂牛奶的百分比是多少？ Protein composes what percentage of full-fat cow's milk? **3-4%** 10-12% 1-2% 69-70%	Sara Stancil, Clemson University, *A study of the frothing capacity of whole milk* Levy M., University of Illinois, *The effects of composition and processing of milk on foam characteristics as measured by steam frothing*

		3. 了解牛奶中的糖分 Understand sugars in milk	乳糖占牛奶中卡路里的40%。 它是由葡萄糖和半乳糖组成的二糖。 人类产生一种分解乳糖的酶，叫作乳糖酶。没有乳糖酶则会发生“乳糖不耐症”。 Lactose accounts for up to 40% of calories in cow’s milk. It is a disaccharide made up of glucose and galactose. Humans produce an enzyme, lactase, that breaks down lactose. Without this a person can become “lactose intolerant”.	**问题 ID 100008892673/100009359969** 选择合适的答案来完成以下句子。 一个人对牛奶有不适症状，这可能是因为这个人无法在体内产生_______，用来分解牛奶中的糖。 半乳糖 乳糖 **<u>乳糖酶</u>** 葡萄糖 Select the correct word to complete the sentence. A person’s intolerance to milk may be because they do not produce the enzyme “________” which breaks down sugar in milk. Galactose Lactose Lactase Glucose	
	3.06.01.02 **牛奶处理** **Milk Processing**	1. 了解巴氏灭菌法和均质化如何影响牛奶的成分 Understand how pasteurization and homogenization impacts the components of milk	牛奶热处理的方法很多，包括巴氏灭菌的各种方法（HTST 以及需要较长时间的较低温度的方法）和诸如 UHT 的较高热处理。 There are numerous methods of milk heat treatment including various methods of pasteurization (HTST as well as lower temperature methods taking longer times) and higher heat treatments like UHT.	**问题 ID 100008892674** HTST（高温/短时）巴氏灭菌对牛奶有什么影响？选择**错误**的答案。 停止牛奶中的酶活性 杀死有害细菌 使少量蛋白质变性 **<u>将脂肪球分散成小分子</u>** What effect does HTST pasteurization (High Temperature /Short Time) **NOT** have on milk?. Stops enzymatic activities in milk Kills harmful bacteria Denatures a small percentage of proteins **<u>Disperses fat globules into small molecules</u>**	Harold McGee, Food & Cooking an Encyclopedia of Kitchen Science, History and Culture, 222-23 Illy & Viani, Espresso Coffee, 312

				问题 ID 100008892676 在哪个过程中，牛奶会在高压下被迫通过小孔，从而减小脂肪小球的大小并将其均匀地分散在其余的牛奶中？ 超高温 **均质化** 巴氏杀菌 During which process is milk forced through small holes at high pressure, reducing the size of the fat globules and dispersing them uniformly through the rest of the milk? UHT **Homogenization** Pasteurization	
				问题 ID 100008892678 牛奶的 HTST（高温/短时间）巴氏灭菌需要什么温度和时间条件？ 95°C 15 分钟 **72°C 15 秒** 63°C 30 分钟 65°C 3 秒 What are the temperature and time conditions required for HTST (High Temperature/Short Time) pasteurization of milk? 95C for 15 minutes **72C for 15 seconds** 63C for 30 minutes 65C for 3 seconds	

3.06.02 **泡沫的产生，质量和稳定性** **Foam Creation, Quality and Stability**		1. 描述蛋白质和脂肪在泡沫产生，稳定性和质地中的作用 Describe the role of protein and fat in foam creation, stability and texture	发泡乳中所含的乳清蛋白（β-乳球蛋白）具有疏水性，在发泡过程中被吸引到空气中，成为气泡的表面活性剂。热量展开并凝结乳清蛋白，以稳定泡沫。 Whey protein (Beta lactoglobulin) involved in foaming milk is hydrophobic and is attracted to air during foaming to become a surfactant for the bubbles. Heat unfolds and coagulates the whey proteins to stabilize the foam. 蒸汽引入空气（通过打发过程）和热量。 Steam introduces air (through a whipping motion) and heat.	**问题 ID 100008892686** 在使牛奶起泡时，哪种蛋白质对泡沫的稳定性影响最大? 酪蛋白蛋白 **乳清蛋白** 酪蛋白和乳清蛋白均无 酪蛋白和乳清蛋白均有 When foaming milk, which type of protein has the greatest contribution to the stability of the foam? Caseins protein **Whey proteins** Neither caseins nor whey proteins Both, casein and whey proteins	Harold McGee, Food & Cooking an Encyclopedia of Kitchen Science, History and Culture, 26-27 Illy & Viani, Espresso Coffee, 312 Münchow, M., Jørgensen, L., Amigo, J., Sørensen, K. and Ipsen, R., *Steam-frothing of milk for coffee*

		2. 了解“失水”如何影响泡沫质量 Understand how “drainage” affects foam quality	从泡沫中排出的液体会导致气泡彼此紧密接触并合并。这导致越来越大的气泡，最终使气泡破裂。 牛奶越热，失水越快。热变性的蛋白质保持稳定，但水分从其中排出。 Liquid draining from foam causes air bubbles to come in close contact with each other and merge. This leads to larger and larger air bubbles, which eventually collapse. The hotter the milk, the faster drainage occurs. Heat denatured proteins remain stable but moisture drains from them.	**问题 ID 100008892688** 选择以下最能使句子完整的答案：提高牛奶加热的温度会使泡沫“排出” ____________________： 更慢且不稳定 **更快且稳定** 更快且不稳定 更慢且稳定 Select the answer below that best completes the sentence. Increasing the temperature that milk is heated to will make foam “drain”_________________________: More slowly and become less stable **Faster and remain stable** Faster and become less stable More slowly and remain stable	

3.06.03 **牛奶的质量和起泡能力** **Milk Quality and Ability to Foam**	**3.06.03.01** **影响牛奶品质的因素** **Factors Affecting Milk Quality**	1. 识别由于牛奶老化或牛奶暴露在阳光和荧光照射下导致的负面影响。以及制作过程失误而导致的泡沫质量下降 Identify deterioration in foam quality caused by aging milk and milk damaged in production and/or route 2. 了解牛奶的最佳储存条件以及不良储存条件的影响 Understand optimum storage conditions for milk and the impact of poor storage conditions	较高的储存温度导致细菌活性增加，脂解和蛋白水解增加。 暴露在阳光下会影响牛奶的风味和品质 暴露在阳光或荧光下会产生一个明显的类似白菜、和烧焦的味道。这是由于维生素和黄色和一些含硫氨基酸的反应。 透明玻璃瓶和塑料容器加上超市的灯光会引起这样的问题。使用避光纸盒可以避免这种状况。 Higher storage temperatures lead to increased bacterial activity, increasing lipolysis and proteolysis. Exposure to light can affect milk flavor and quality The light can cause oxidation of the milk, where oxygen is added into the protein cell and riboflavin and amino acids are produced. This can impair the flavor.	**问题 ID 100008892690** 暴露在阳光或荧光灯照射下会对牛奶产生什麽负面影响? 脂解酸败 **核黄素氧化** 焦糖化 金属氧化 What negative impact will exposure to sunlight or fluorescent lighting have on milk? Lipolytic rancidity **Riboflavin oxidation** Caramelization Metallic oxidation	Harold McGee, Food & Cooking an Encyclopedia of Kitchen Science, History and Culture, 21

	3.06.03.02 **蛋白水解和脂解** **Proteolysis and Lipolysis**	1. 识别并理解为什么受脂解和蛋白水解作用的牛奶不会起泡沫。Identify and understand why milk will not foam when affected by lipolysis and proteolysis. 2. 描述引起脂解和蛋白水解的过程 Describe the processes that cause lipolysis and proteolysis	**脂解** **Lipolysis** • 脂肪酶对乳脂的酶促降解 • 结果，气泡变得不稳定。 • 脂肪分解还会产生腐臭的味道和气味 • Enzymatic degradation of milk fat by the enzyme lipase • As a result, the air bubbles become unstable. • Lipolysis can also create a rancid taste and smell **蛋白水解** **Proteolysis** • 细菌酶（蛋白酶）将蛋白质分解为肽和氨基酸。 • 因此，牛奶存放越久，酸度越高，就会引起凝乳和乳清。 • Bacterial enzymes (proteases) break proteins into peptides and amino acids. • Hence the older the milk the more acidic it becomes causing curds and whey.	**问题 ID 100008892692** 牛奶中的脂肪分解为游离脂肪酸和甘油三酸酯的过程被称为： **<u>脂解</u>** **蛋白**水解 水解 均质 The breakdown of the fats in milk, into free fatty acids and triglyceride, is known as: **<u>Lipolysis</u>** Proteolysis Hydrolysis Homogenization **问题 ID 100008892694** 以下哪个陈述最能说明蛋白水解作用? **<u>酶分解蛋白质;</u>** 脂肪在整个牛奶中的分布均匀; 牛奶中脂肪的分解; 稳定牛奶泡沫的蛋白质。 Which statement below best describes proteolysis? **<u>Enzymes breaking down proteins.</u>** The distribution of fats evenly throughout the milk. The breakdown of fats within the milk. Proteins stabilizing milk foams.	http://www.milkfacts.info/Milk%20Composition/Protein.htm Harold McGee, Food & Cooking an Encyclopedia of Kitchen Science, History and Culture, 17-21

	3.06.03.03 咖啡酸度对牛奶的影响 Coffee Acidity's Effect on Milk	1. 了解咖啡是酸性的，会引起与牛奶蛋白质的反应 Understand that coffee is acidic and can cause a reaction with milk proteins	牛奶中有数百个被 Kappa-酪蛋白封端的蛋白质小簇。 钙起到胶水的作用，有助于将酪蛋白结合成胶束。 较高的酸度（pH 值 4.7 或更酸）可溶解钙键并中和另一端的酪蛋白–简单地说，高酸度会使牛奶变成凝乳和乳清。 牛奶的正常 pH 值在 6.5 – 6.7 之间（略带酸性） Milk has hundreds of small clusters of proteins capped by Kappa-caseins. Calcium acts as a glue to help bind the casein proteins into micelles. Higher acidity, ph 4.7 or more, dissolves the calcium bonds and neutralizes the capping caseins – Simply, high acidity turns milk into curds and whey. The normal ph value of milk is between 6.5 – 6.7 (slightly acidic)	**授课，不考。Taught, not tested.**	

	3.06.03.04 **热量对牛奶的影响** **Heat's Effect on Milk**	1. 确定过热使牛奶蛋白变性的方式和原因，从而使风味变差 Identify how and why excessive heat denatures milk protein thereby deteriorating flavor	热会使蛋白质变性，从而影响风味并产生硫化氢，产生"蛋"样的气味。在 HTST 巴氏灭菌中，在 72°C 的温度下，高达 10%的蛋白质已经变性。 Heat denatures protein affecting flavor and creating hydrogen sulfide, giving the "egg' like smell. In HTST pasteurization, at temperatures of 72C, up to 10% of proteins are already denatured.	**问题 ID 100008892695** 过度加热牛奶会导致"蛋白质变性"。过热会对牛奶的味道和气味产生什么影响? 气味和风味将是最甜，很奶油般的。 **气味和味道会有鸡蛋般的气味和煮熟的味道。** 蒸汽加热后，气味和风味不会改变。 Excessively heating milk can lead to "denatured proteins". What effect would excessive heat have on the flavor and smell of milk? The smell and flavor will be at its sweetest and most creamy. **The smell and flavor will have an egg like smell and cooked flavor.** The smell and flavor do not change when steamed cool to hot.	http://www.milkfacts.info/Milk%20Composition/Protein.htm
				问题 ID 100008892697 不应将牛奶加热到沸点的主要原因是什么? **蛋白质将被变性，影响风味;** 牛奶将开始失去其甜味; 脂肪将开始分解，防止起泡; 蛋白质将开始凝结。 What is the main reason why milk should not be heated to its boiling point? **Proteins will be denatured affecting flavor.** Milk will start to lose its sweetness. The fats will start to break down preventing foaming. Proteins will start to curdle.	

				问题 ID 100008892698 以下陈述是对还是错？当使牛奶起泡时，过多的热量不会影响脂肪，因为与蛋白质相比，脂肪更稠密，分子更大。 对 **错** Is the following statement true or false? When foaming milk, excessive heat does **NOT** affect fats, as they are more dense and larger molecules compared with proteins. True **False**	
3.06.04 **SCA 泡沫标准** **SCA Foam Standards**		1. 识别并制作始终如一的高质量微泡沫（SCA 咖啡师泡沫标准的最高水平） Recognize and produce consistently high-quality micro- foam (Highest level on SCA Barista Foam Standards)		**在实践考试中考核。** **Tested on Practical Exam.**	SCA, *Barista Foam Standards* WBC, *Rules and regulations* WLAC, Rules and regulations
3.06.05 **SCA 拿铁艺术标准-直接拉花** **SCA Latte Art Standards - Free Pour**		1. 按照 SCA 拿铁艺术标准展示两种拉花图案 Demonstrate ability to free pour two latte art patterns to highest SCA Latte Art Standards 2. 展示出能够制作一致拉花图案的能力 Demonstrate ability to produce, in a pair, consistent latter art patterns		**在实践考试中考核。Tested on Practical Exam.**	SCA, *Barista Latte Art Standard*

3.07|栏目|水质 | Section | WATER QUALITY

2 主题

主题 Topic	**子主题** Sub-Topic	**目标** Objectives	**AST 注意事项** AST Notes	**在线笔试题** Online Written Exam Questions	**参考资料** References
3.07.01 **测量** **Measurements**	**3.07.01.01** **总溶解固体** **(TDS)** **Total Dissolved Solids (TDS)**	1. 测量水的 TDS 并了解结果将如何影响冲煮 Measure the TDS of water and understand how the results will influence brewing	电导率仪是测量水中 TDS 的常用工具。 Conductivity meter is the common tool to measure TDS in water.	**问题 ID 100008892700** 使用什么设备/工具测量水中的总溶解固体（TDS)? 折光仪 **电导率仪** 咖啡测试仪 比重计 What equipment/tool is used to measure Total Dissolved Solids (TDS) in water? Refractometer **Conductivity meter** Coffee test meter Hydrometer	SCA, Water Quality Handbook, 54, 81 Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 389-390
				问题 ID 100008892702 为了达到最佳冲煮，建议的水 TDS 范围是多少? 70 - 100 ppm (mg/l) **125 - 175 ppm (mg/l)** 140 - 180 ppm (mg/l) 40 - 75 ppm (mg/l) What is the recommended range of water TDS to achieve optimal brews?	

	3.07.01.02 **碱度** **Alkalinity**	1. 测量碱度并了解结果将如何影响冲煮 Measure alkalinity and understand how the results will influence brewing	回顾碱度及其对浓缩咖啡的影响。 演示并让学习者使用碱度滴剂套件。 Review alkalinity and its effect on espresso. Demonstrate and have learners use alkalinity drops kit.	**问题 ID 100008892703** 为了达到最佳的冲煮效果，水的碱度建议范围是多少? **40-75 ppm $CaCO_3$ 碱度 alkalinity** 51-175 ppm $CaCO_3$ 碱度 alkalinity 30-100 ppm $CaCO_3$ 碱度 alkalinity 75-175 ppm $CaCO_3$ 碱度 alkalinity What is the recommended range of water alkalinity to achieve optimal brews?	SCA, Water Quality Handbook, 12-13 Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 383
				问题 ID 100008892704 增加冲煮水的碱度通常会导致以下哪种情况发生? 增加冲煮咖啡中酸度的感知 **增加设备上形成水垢的风险** 降低设备上形成水垢的风险 Increasing the alkalinity of the brewing water will cause which of the following to typically occur? Increases the perception of acidity in the brewed coffee **Increases the risk of limescale forming on equipment** Lowers the risk of limescale forming on equipment	

				问题 ID 100008892705 碱度为 10 ppm $CaCO_3$的水将如何影响饮料的口味，并会对浓缩咖啡机造成风险? **该饮料可能会变酸，并且有很高的腐蚀风险.** 该饮料可能会出现粉渣口感并且尝起来平乏，并且 有较高的水垢风险。 该饮料可能会出现粉渣口感并且尝起来平乏，但没有技术风险。 该饮料可能味道平衡，并且没有技术风险。 How will water with alkalinity of 10 ppm $CaCO_3$ likely affect the taste of the beverage, and would it pose risks to the espresso machine? **The beverage will likely taste sour, and there is a high risk of corrosion,** The beverage will likely taste chalky and flat, and there is a higher risk of limescale. The beverage will likely taste chalky and flat, but there is no technical risk. The beverage will likely taste balanced, and there is no technical risk.	
	3.07.01.03 **总硬度（TH）** **Total Hardness (TH)**	1. 测量总硬度（TH）并了解结果将如何影响冲煮 Measure total hardness (TH) and understand how the results will influence brewing	回顾总硬度（TH）及其对浓缩咖啡的影响， 演示并让学习者使用总硬度 TH 滴剂套件。 Review total hardness (TH) and its effect on espresso. Demonstrate and have learners use total hardness TH drops kit.	**问题 ID 100008892706** 选择最能使句子完整的答案。总硬度为 300 ppm $CaCO_3$___________。 和碱度为 40ppm 的水会引起水垢风险 **和碱度为 200ppm $CaCO_3$会引起水垢风险** 和碱度为 200ppm $CaCO_3$会引起腐蚀风险 和碱度为 150ppm 會引起腐蚀风险 Select the answer that best completes the sentence. Water with a total hardness of 300 ppm $CaCO_3$ ___________. and alkalinity of 40ppm poses a limescale risk **and alkalinity of 200ppm $CaCO_3$ poses a limescale risk** and alkalinity of 200ppm $CaCO_3$ poses a corrosion risk and alkalinity of 150ppm poses a corrosion risk	SCA, Water Quality Handbook, 12-13, 56 Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 384-390

				问题 ID 100008892708 关于水垢的哪项陈述**不**正确？ **水垢是碳酸镁的沉淀和碳酸钙的溶解导致的。** 严重的水垢会导致浓缩咖啡机的特定金属部件过热。 由于水垢引起的管道收缩会增加水流阻力和泵过载。 仅 1.0 毫米厚的石灰垢层可将锅炉中热交换器的传热系数降低多达 80%。 Which statement about limescale is **NOT** true? **Limescale is a precipitation of magnesium carbonate and the dissolving of calcium carbonate.** Severe limescale can lead to overheating specific metal parts of the espresso machine. Constriction of pipes, due to limescale, increases water flow resistance and pump overloading. Limescale coating of only 1.0mm thickness can reduce the heat transfer coefficient of heat exchanger in a boiler by up to 80%.	
				问题 ID 100008892709 达到最佳冲煮的水总硬度的建议范围是多少？ 40-75 ppm 碳酸钙总硬度 **50-175 ppm 碳酸钙总硬度** 30-100 ppm 碳酸钙总硬度 75-175 ppm 碳酸钙总硬度 What is the recommended range of water total hardness to achieve optimal brews? 40 - 75 ppm $CaCO_3$ total hardness **50 - 175 ppm $CaCO_3$ total hardness** 30 - 100 ppm $CaCO_3$ total hardness 75 - 175 ppm $CaCO_3$ total hardness	SCA, Water Quality Handbook, 12-13

				问题 ID 100008892711 总硬度为 220 ppm $CaCO_3$ 的水将如何影响饮料的味道? **咖啡会变浓郁。** 咖啡口感变粗糙。 咖啡的味道会变弱。 咖啡会变平乏。 How will water with a total hardness of 220 ppm $CaCO_3$ likely affect the taste of the beverage? **Coffee will taste heavy.** Coffee will taste chalky. Coffee will taste weak. Coffee will taste flat.	SCA, Water Quality Handbook, 12-13
	3.07.01.04 pH	1. 测量 pH 值并了解结果将如何影响冲煮 Measure pH and understand how the results will influence brewing	回顾 pH 值及其对浓缩咖啡的影响， 演示并让学习者使用数字仪表/条。 Review pH and its effect on espresso. Demonstrate and have learners use digital meter / or strips.	**问题 ID 100008892713** 根据 SCA 水质手册，咖啡冲煮水的建议 pH 范围是多少? According to the SCA Water Quality Handbook, what is the recommended pH range for coffee brewing water? 4 - 5 3 - 9 7 - 9 **6 - 8**	SCA, Water Quality Handbook, 11, 41-43, 75

				问题 ID 100008892715 pH 值低于建议范围的水最有可能带来什么风险? 水垢 **腐蚀** 浓缩咖啡机上的水管堵塞 浓缩咖啡机上的电子组件过热 What is the most likely risk associated with water that has a pH lower than the recommended range? Limescale **Corrosion** Blockage of water pipes on espresso machine Overheating of electrical components on espresso machine	
3.07.02 **水过滤** **Water Filtration**	**3.07.02.01** **水质测试与过滤系统** **Water Testing and Filtration System**	1. 根据测试结果认识到水过滤的必要性 Recognize the need for water filtration based on testing results		**问题 ID 100008892717** 水测试显示碱度为 225 ppm CaCO3。下一步需要做什么? **使用脱碳剂处理将 $CaCO_3$ 的数值降低至 40-75 ppm** 使用碳化器处理将 $CaCO_3$ 的数值提高至 250-300 ppm 没什么，因为它在建议的范围内 Water testing shows alkalinity of 225 ppm $CaCO_3$. What is a necessary next step? **Use a decarbonizer treatment to bring level down to 40 - 75 ppm $CaCO_3$** Use a carbonizer treatment to bring level up to 250 - 300 ppm $CaCO_3$ Nothing, as it is in the recommended range	SCA, Water Quality Handbook, 55-59 Illy & Viani, Espresso Coffee, 280-282

				问题 ID 100008892718 如果在意式浓缩咖啡机的可见区域发现水垢，应进行哪种水质测试以确定可能的原因？ 总硬度 pH 碱度 **以上所有** When limescale is noticed in visible areas of the espresso machine, which water test should be done to identify the possible causes? Total hardness pH Alkalinity **All of the above**	
				问题 ID 100008892720 是否所有水都需要软化？ 是，硬度是设备和咖啡萃取的问题所在 **否，首先测量水并评估是否需要水处理** **是**，总硬度和碳酸盐（缓冲液）应始终降低到最低限度 是，设备供应商将不接受安装在设备上的过滤器出现任何不使用软化剂的保修要求 Does all water need to be softened? YES, hardness is a problem for equipment and coffee extraction **NO, first measure the water and evaluate if any water treatment is required** YES, total hardness and carbonates (buffer) should always be reduced to a minimum YES, equipment suppliers will not accept any warranty claims without softener filter installed on the equipment	

	3.07.02.02 **水过滤系统的功能和类型** **Function and Types of Water Filtration Systems**	1. 描述咖啡馆常用的水过滤系统的功能 Describe the function of common water filtration systems used in cafes 2. 建议适合不同水成分的过滤 Recommend suitable filtration for different water composition	解释以下滤水器系统如何工作的一般方面: ● 碳过滤器 ● 软化剂（钠） ● 脱碳机 ● 脱盐剂（反渗透） Explain the general aspects of how the following water filter systems work: ● Carbon filters ● Softener (sodium) ● Decarboniser ● Demineraliser (Reverse Osmosis)	**问题 ID 100008892721** 如果水中有微量氯，且总硬度为 75 ppm，碱度为 50ppm，那么哪种过滤器系统被认为是合适的? **碳过滤器** 带 10%旁路的反渗透 碳过滤器的钠软化 剂脱碳器（DC） What filter system could be considered appropriate if the water has chlorine traces and a total hardness of 75 ppm and an alkalinity of 50ppm? **Carbon filter** Reverse osmosis with 10% bypass Sodium softener with carbon filter Decarboniser (DC)	Rao, The Professional Barista's Handbook, 84-85 SCA, Water Quality Handbook, 27-37 Folmer, Britta, and Imre Blank. The Craft and Science of Coffee, 391, 395
				问题 ID 100008892723 在不影响碱度的前提下，最常见的降低总硬度的水处理方法是? 反渗透（RO） 脱盐处理（DM） **软化剂处理** 脱碳剂（DC） Which is the most common water treatment to reduce total hardness without affecting alkalinity? Reverse osmosis (RO) Demineralising treatment (DM) **Softener treatment** Decarboniser (DC)	

				问题 ID 100008892724 如何在反渗透（RO）水处理中使用旁路? 防止未经处理的水进入反渗透系统。 处理过的水以相反的方向穿过反渗透膜 **反渗透前的水再矿化以达到所需的 TDS** 一定百分比的水绕过膜并用于冲洗杂质从系统 How is bypass used in reverse osmosis (RO) water treatment? Untreated water is prevented from entering the RO system Treated water passes through the RO membrane in reverse direction **Pre-RO water is used to re-mineralize the RO water to achieve a desired TDS** A percentage of water bypasses the membrane and is used to flush impurities from the system	

3.08 |栏目| 简单的金融概念 SIMPLE FINANCIALS CONCEPTS

1 个主题

主题 Topic	**目标 Objectives**	**AST 注意事项 AST Notes**	**在线笔试题 Online Written Exam Questions**	**参考资料 References**
3.08.01 **成本与商品** **Cost and Goods**	使用标准技术（毛利润，成本加成等）分析饮料样本价格，以了解如何使用定价模型 Understand how costs and sales volumes are used in determining the price of menu items	**销售货物的成本 （COGS）** 是指公司销售的商品的产生成本。此金额包括直接用于创造产品的材料和人工成本。它不包括间接费用，如分销成本和销售队伍成本。 **Cost of goods** sold (**COGS**) refers to the direct costs of producing the **goods** sold by a company. This amount includes the **cost** of the materials and labor directly used to create the good. It excludes indirect expenses, such as distribution costs and sales force costs. **边际收益 （CM），** 是单个单位减去可变成本的销售价格，用于生成该单位。生成的金额是单位为帮助支付固定成本而产生的收入金额。请注意，收益利润率与毛利率不一样。毛利率通过从部门或公司特定时期的收入中减去直接成本来确定。毛利率是指企业的业绩，而收益利润率是指一种产品的潜力。 **Contribution margin** (CM), is the selling price a single unit minus variable cost to produce that single unit. The resulting amount is the amount of revenue that unit generates to help cover fixed costs. Note that contribution margin is not the same as gross profit margin. Gross profit margin is determined by subtracting direct costs from revenue over a specific period of time for a department or company. Gross profit margin refers to the performance of an business while contribution margin refers to the potential of one product. **收支平衡点**可以定义为总成本（费用）和总销售额（收入）相等的点。**Break-even point** can be defined as a point where total costs (expenses) and total sales (revenue) are equal.	**该主题有一个活动。** **This topic has an activity.** **见 8。附录-活动准备指南 See 8. Appendix - Activities Preparation Guide**	Colin Harmon, What I Know About Running Coffee Shops, 203-209

5. SCA 基本培训文件 Essential SCA Training Documents

- SCA Barista Foam Standards
- SCA Latte Art Standards
- SCA Barista Drink Standards
- SCA Coffee Taster’s Flavor Wheel (English)
- SCA Water Chart
- SCA Barista Professional Mandatory Activities Worksheet

所有文档都可以在 AST 门户网站的“课程和考试/咖啡师技能”下找到。All documents are available at the AST Portal under Curriculum and Exams/Barista Skills.

6. 所需设备和用品清单 Required Equipment and Supplies List

可在 AST Portal 的“资源/场地要求”下找到。Available at the AST Portal under Resources/Venue Requirements.

SCA 美国或英国商店中可以购买的物品均会提供链接。Any items available in the SCA US or UK store are noted and a link directly to the store is provided.

7. 参考书目 Bibliography

- Stephenson, T., *The curious barista's guide to coffee*. London: Ryland Peters & Small, 2019.
- Folmer, Britta, and Imre Blank. *The Craft and Science of Coffee*. Boston, MA: Elsevier, 2017.
- Illy, A., Viani, R. and Suggi Liverani, F. *Espresso coffee*. Amsterdam: Elsevier Academic Press, 2005.
- Roast Magazine, *The Book Of Roast*. Portland, Oregon: JC Publishing, Inc., 2017.
- Sage, E.. SCA *The Coffee Biology Field Glossary*. 2nd ed., 2017.
- World Coffee Research.org, *F1 Hybrids Explainer,* 2020. [online] Available at: https://worldcoffeeresearch.org/work/next-generation-f1-hybrids/f1-hybrids-explainer/
- PT's Coffee.. *Experimental Coffee Processing Methods,* Explained, 2020. [online] Available at:
- https://www.ptscoffee.com/blogs/news/experimental-coffee-processing-methods-explained
- Kornman, C. *A., Guide To Carbonic Maceration And Anaerobic Fermentation In Coffee*, 2020 [online] Daily Coffee News by Roast Magazine. Available at: https://dailycoffeenews.com/2019/06/03/a-guide-to-carbonic-maceration-and-anaerobic-fermentation-in-coffee/
- Clarke, R. and Vitzhum, R. *Coffee Recent Developments*. Oxford: Blackwell Science (2001).
- Cottrell, C., 2013. *Barista Bible*. Willston, Qld.: Coffee Education Network, 2013.

- Scanews.coffee. *Milk Foam: Creating Texture And Stability | Specialty Coffee Association News*, 2020. [online] Available at: https://scanews.coffee/2014/09/15/milk-foam-creating-texture-and-stability/
- Sara Stancil, Clemson University, A study of the frothing capacity of whole milk, 2013
- Levy, M., *The Effects Of Composition And Processing Of Milk On Foam Characteristics As Measured By Steam Frothing*. [online] Citeseerx.ist.psu.edu. 2021. Available at: http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.497.7595&rep=rep1&type=pdf
- Münchow, M., Jørgensen, L., Amigo, J., Sørensen, K. and Ipsen, R., . Steam-frothing of milk for coffee: Evaluation for foam properties using video analysis and feature extraction. *International Dairy Journal*, 2015.
- Marco Wellinger, Samo Smrke & Chahan Yeretzian, *The Water Chart*, SCAE, 2016.
- Verein Deutscher Ingenieure, *Analytical Proceedings guidelines*,1988.
- Harmon, C., n.d. *What I Know About Running Coffee Shops*.
- World Latte Art Championship, Rules & Regulations, http://www.worldlatteart.org/rules/, 2019.
- Lingle, T. *The coffee brewing handbook*. Speciality Coffee Association of America, 2011.
- Marco Wellinger, Samo Smrke & Chahan Yeretzian, *The Water Chart*, SCAE, 2016.
- Colonna-Dashwood, Maxwell, and Christopher H. Hendon. *Water for Coffee: Science, Story, Manual*. Bath: Maxwell Colonna-Dashwood and Christopher H. Hendon, 2015.
- Milk Facts.Info. *Milk Protein* |, 2020. [online] Available at: http://www.milkfacts.info/Milk%20Composition/Protein.htm
- Sivetz, M. and Desrosier, N., *Coffee Technology*. Westport, Conn.: AVI Pub. Co.
- McGee, Harold. *McGee on Food & Cooking: an Encyclopedia of Kitchen Science, History and Culture*. London: Hodder & Stoughton, 2004.
- Fury, M., *To Freeze Or Not To Freeze, That Is The Coffee Question* - Perfect Daily Grind., 2020. [online] Available at: https://www.perfectdailygrind.com/2015/10/to-freeze-or-not-to-freeze-that-is-the-coffee-question/
- Dockrill, P., *Want To Drink Better-Tasting Coffee? Freeze Your Beans, Say Scientists*. [online] Science Alert, 2020. Available at: https://www.sciencealert.com/want-to-drink-better-tasting-coffee-freeze-your-beans-say-scientists
- Huladaddy.com. *Freezing Coffee Beans Keeps Them Fresh | Huladaddy.Com*, 2020. [online] Available at: https://www.huladaddy.com/articles/freeze-your-beans.htm
- Compoundcoffee.com. *Does Grinding Frozen Coffee Beans Impact Brewing & Taste?*, 2020. [online]. Available at: https://compoundcoffee.com/experiments/25_Does-Grinding-Frozen-Coffee-Bean-Impact-Taste
- *Barista drink standard,* [English], London: Specialty Coffee Association, 2019.
- Speaker Deck., *Barista Camp 2015 | Christian Klatt — Heating (In) Grinders*, 2020. [online] Available at: https://speakerdeck.com/bge/barista-camp-2015-christian-klatt-heating-in-grinders?slide=21
- Coffee for us. *10 Best Espresso Tampers In 2020 Reviewed | Coffee For Us*, 2020. [online] Available at: https://coffeforus.com/best-espresso-tampers/

- Socraticcoffee.com. *Comparing The Impact Of Tamper On Extraction – Socratic Coffee*, 2020 [online] Available at: http://socraticcoffee.com/2015/08/comparing-the-impact-of-tamper-on-tds/
- *Coffee Freshness Handbook,* [English], London: Specialty Coffee Association, 2016.
- Hoffmann, J. *The world atlas of coffee*. London: Mitchell Beazley, 2018.
- Rao, S. *The professional barista's handbook*. USA: Scott Rao, 2008.

8 - 附录-活动准备指南 Appendix - Activities Preparation Guide

- 本课程的活动包含在咖啡师专业必修活动工作表中。**每个学员必须接收并完成工作表。**工作表位于"咖啡师下载包"中，位于专业指南文件夹中。Activities for this course are contained in the Barista Professional Mandatory Activities Worksheet. **Each learner must receive and complete the worksheet**. The worksheet is in the Barista download pack, in the Professional Guidebook folder.
- AST 还应查阅以下活动说明以了解更多信息。ASTs should also consult the activity instructions below for any additional information.
- 这些活动是课程的必修部分，但不会通过考试进行正式测试。These activities are a mandatory part of the course but will not be tested formally by examination.

<table>
<tr><th>编码 Code</th><th>使用说明 Instructions</th></tr>
<tr><td>3.03.03
填压和布粉评估 Tamper and Distribution Evaluation</td><td>在开展此活动之前，AST 应仔细设计和校准浓缩咖啡冲煮配方 Prior to this activity, the AST should carefully design and calibrate an espresso brew recipe.</td></tr>
<tr><td>3.07.01
（水质）测量和品尝
(Water Quality) Measurements and Tasting</td><td>没有额外的指示 No additional instructions</td></tr>
<tr><td>3.08 简单财务概念
Simple Financials Concepts</td><td>请注意下面标注红色的答案 Note the correct answers below in red:
<table>
<tr><th>可见支出 Available Costs</th><th>(€)</th><th>已售部分支出（可变支出）Cost of Goods Sold (variable costs)</th><th>(€)</th></tr>
<tr><td>咖啡熟豆 Roasted coffee</td><td>0.4</td><td>咖啡熟豆 Roasted coffee</td><td>.4</td></tr>
<tr><td>电力等 Electricity</td><td>1.2</td><td></td><td></td></tr>
<tr><td>纸杯 Paper cup</td><td>0.06</td><td>纸杯 Paper cup</td><td>.06</td></tr>
<tr><td>杯盖 Lid</td><td>0.02</td><td>杯盖 Lid</td><td>.02</td></tr>
<tr><td>房租 Rent</td><td>2</td><td></td><td></td></tr>
<tr><td>牛奶 Milk</td><td>0.4</td><td>牛奶 Milk</td><td>.4</td></tr>
<tr><td>直接劳务 Direct Labor</td><td>.3</td><td>直接劳务 Direct labor</td><td>.3</td></tr>
</table></td></tr>
</table>

保险 Insurance	.05		
会计费用 Accounting Fees	.05		
市场宣传 Marketing	.5		
		总产品费用 Cost of Goods Total:	1.18

*税款是代表政府征收的，不是收入（销售）的一部分，因此无需纳税即可计算全部税款（如增值税等）。

*Tax is collected on behalf of the GOVERNMENT and is NOT part of the Revenue (Sales) so calculate all without Tax (e.g. VAT etc.).

销售商品的成本占销售价格的百分比是多少?

The Cost of Goods Sold is what percentage of the selling price? 23.6%

单价 Unit Price €5 - __€1.18____ = _____€3.82_______

(销售货物成本 COGS) (边际收益 Contribution Margin)

固定支出 Fixed Costs of €5,000 / ______€3.82_______ = _______1,309___________

(边际收益 Contribution Margin) (Break Even Sales Volume)